



Note: Pharma code & Product name position and direction is not fixed and may vary based on the printer's requirement/ Machine suitability.

360 mm

240 mm

Twenty percent of the patients experienced a drop in their platelet count below 100,000 cells/mm3 at least once while on treatment; 7% had a platelet count <50,000 cells/mm3 at the time of their worst nadir. Bleeding episodes were reported in 4% of all courses and by 14% of all patients, but most of the hemorrhagic episodes were localized and the frequency of these events was unrelated to the Paclitaxel dose and schedule.

Hypersensitivity Reactions (HSR): All patients received premedication prior to Paclitaxel therapy. The frequency and severity of HSR were not affected by the dose or schedule of Paclitaxel administration. The most frequent symptoms observed during these severe reactions were dyspnea, flushing, chest pain and tachycardia. Abdominal pain, pain in the extremities, and hypertension are also noted. Minor hypersensitivity reactions, mainly flushing and rash, did not require therapeutic intervention nor did they prevent continuation of Paclitaxel therapy.

Cardiovascular: Hypotension, during the first 3 hours of infusion, occurred in 12% of all patients and 3% of all courses administered. Bradycardia, during the first 3 hours of infusion, occurred in 3% of all patients and 1% of all courses.

Severe cardiac conduction abnormalities have been reported in <1% of patients during Paclitaxel therapy. If patients develop significant conduction abnormalities during Paclitaxel administration, appropriate therapy should be administered and continuous electrocardiographic monitoring should be performed during subsequent therapy with Paclitaxel.

Neurologic: In general, the frequency and severity of neurologic manifestations were dose-dependent in patients receiving single-agent Paclitaxel.

The frequency of peripheral neuropathy increased with cumulative dose. Paresthesia commonly occurs in the form of hyperesthesia. Peripheral neuropathy was the cause of Paclitaxel discontinuation in 1% of all patients. Sensory symptoms have usually improved or resolved within several months of Paclitaxel discontinuation. Pre-existing neuropathies resulting from prior therapies are not a contraindication for Paclitaxel therapy.

Rare reports in the literature of abnormal visual evoked potentials in patients have suggested persistent optic nerve damage.

Gastrointestinal (GI) Toxicity: Mild to moderate nausea/vomiting, diarrhea and mucositis were reported very commonly by all patients. Mucositis was scheduled dependent and occurred more frequently with the 24-hour than with the 3-hour infusion.

Rare reports of neutropenic enterocolitis (typhlitis), despite the coadministration of G-CSF, were observed in patients treated with Paclitaxel alone and in combination with other chemotherapeutic agents.

Injection Site Reaction: During intravenous administration, injection site reactions were usually mild and consisted of localized edema, pain, erythema, tenderness, and induration; on occasion, extravasation can result in cellulitis. Skin sloughing and/or peeling has been reported, sometimes related to extravasation. Skin discoloration may also occur.

These reactions have been observed more frequently with the 24-hour infusion than with the 3-hour infusion. In some cases the onset of the injection site reaction either occurred during a prolonged infusion or was delayed by a week to ten days.

Unless otherwise noted, the table below lists undesirable effects regardless of severity associated with the administration of single agent Paclitaxel (812 patients treated in clinical studies) or as reported in the postmarketing surveillance* of Paclitaxel.

The frequency of undesirable effects listed below is defined using the following convention:

Very common; common; uncommon ; rare ; very rare.

System Organ Class	Frequency /Adverse Reactions
Infections and infestations:	Very common: infection Uncommon: septic shock Rare*: sepsis, pneumonia
Blood and the lymphatic system disorders:	Very common: myelosuppression, neutropenia, anemia, thrombocytopenia, leucopenia, fever, bleeding Rare: febrile neutropenia Very rare*: acute myeloid leukemia, myelodysplastic syndrome
Immune system disorders:	Very common: minor hypersensitivity reactions (mainly flushing and rash) Uncommon: significant hypersensitivity reactions requiring therapy (e.g., hypotension, angioneurotic edema, respiratory distress, general zed urticaria, edema, back pain, chills) Rare: anaphylactic reactions (with fatal outcome) Very rare*: anaphylactic shock
Metabolism and nutrition disorders:	Very rare*: anorexia
Psychiatric disorders:	Very rare*: confusional state
Nervous system disorders:	Very common: neurotoxicity (mainly: peripheral neuropathy) Rare: motor neuropathy (with resultant minor distal weakness) Very rare*: autonomic neuropathy (resulting in paralytic ileus and orthostatic hypotension) grand mal seizures, convulsions, cerephalopathy, dizziness, headache, ataxia
Eye disorders:	Very rare 1: reversible optic nerve and/or visual disturbances (scintillating scotomata), particularly in patients who have received higher doses than recommended photopsia, visual floaters.
Ear and labyrinth disorders:	Very rare*: hearing loss, tinnitus, vertigo, ototoxicity
Cardiac disorders:	Very common: abnormal ECG Common: bradycardia Uncommon: cardiomyopathy, asymptomatic ventricular tachycardia, tachycardia with bigeminy, AV block and syncope, myocardial infarction. Very rare* : atrial fibrillation, supraventricular tachycardia
Vascular disorders:	Very common hypotension Uncommon: hypertension, thrombosis, thrombophlebitis Very rare* : shock
Respiratory, thoracic and mediastinal disorders:	Rare*: dyspnea, pleural effusion, respiratory failure, interstitial pneumonia, lung fibrosis, pulmonary embolism Very rare*: cough
Gastrointestinal disorders:	Very common: nausea, vomiting, diarrhea, mucosal inflammation Rare*: bowel obstruction, bowel perforation, ischemic colitis, pancreatitis Very rare*: mesenteric thrombosis, pseudomembranous colitis, esophagitis, constipation, ascites
Hepatobiliary disorders:	Very rare 1: hepatic necrosis (with fatal outcome), hepatic encephalopathy (with fatal outcome)
Skin and subcutaneous tissue disorders:	Very common: alopecia Common: transient and mild nail skin changes Rare*: pruritus, rash, erythema, phlebitis, cellulitis, skin exfoliation, necrosis and fibrosis, radiation recall Very rare*: Stevens-Johnson syndrome, epidermal necrolysis, erythema multiforme, exfoliative dermatitis, urticaria, onycholysis (patients on therapy should wear sun protection on hands and feet)
Musculoskeletal and connective tissue disorders:	Very common arthralgia, myalgia
General disorders and administration site conditions:	Common: injection site reactions (including localized edema, pain, erythema, induration, on occasion extravasation can result in cellulitis) Rare* : asthenia, malaise, pyrexia, dehydration, edema
Investigations:	Common: severe elevation in AST (SGOT): severe elevation in alkaline phosphatase Uncommon: severe elevation in bilirubin Rare*: Increase in fcllood creatinine

Adverse event experiences from studies with combination treatment

The following discussion refers to previously untreated patients with ovarian carcinoma or NSCLC who received Paclitaxel in combination with cisplatin, patients with inoperable NSCLC who received single-agent Paclitaxel in combination with Best Supportive Care, patients with breast cancer who received Paclitaxel after doxorubicin/cyclophosphamide in the adjuvant setting, patients with metastatic breast cancer who received Paclitaxel as first-line therapy with trastuzumab, and patients with AIDS-related Kaposi's sarcoma. In addition, rare events that have been reported from postmarketing experience or

from other clinical studies are described.

PaclitaxelH- Cisplatin

When administered as a three hour infusion for the first-line chemotherapy of ovarian cancer, neurotoxicity, arthralgia/myalgia, and hypersensitivity were reported as more frequent and severe by patients treated with Paclitaxel followed by cisplatin than patients treated with cyclophosphamide followed by cisplatin. Myelosuppression appeared to be less frequent and severe with Paclitaxel as a three-hour infusion followed by cisplatin compared with cyclophosphamide followed by cisplatin.

Cross-study comparison of neurotoxicity in CA 139-209 and CA139-022 suggests that when Paclitaxel is given in combinations with cisplatin 75 mg/m², the incidence of severe neurotoxicity is more common at a Paclitaxel dose of 175 mg/m² given by 3-hour infusion (21%) than at a dose of 135 mg/m² given by 24-hour infusion (3%)

Patients treated with Paclitaxel and cisplatin may have an increased risk of renal failure during the combination therapy of paclitaxel and cisplatin in gynecological cancers as compared to cisplatin alone.

Paclitaxel + Trastuzumab

When Paclitaxel was administered as a 3-hour infusion in combination with trastuzumab for the first-line treatment of patients with metastatic breast cancer, the following events (regardless of relationship to Paclitaxel or trastuzumab) were reported more frequently than with single agent Paclitaxel: heart failure, infection, chills, fever, cough, rash, arthralgia, tachycardia, diarrhea, hypertonia, epistaxis, acne, herpes simplex, accidental injury, insomnia, rhinitis, sinusitis, and injection site reaction. Some of these frequency differences may be due to the increased number and duration of treatments with Paclitaxel/ trastuzumab combination vs single agent Paclitaxel. Severe events were reported at similar rates for Paclitaxel/trastuzumab and single agent Paclitaxel.

Administration of trastuzumab in combination with Paclitaxel in patients previously treated with anthracyclines resulted in an increased frequency and severity of cardiac dysfunction in comparison with patients treated with Paclitaxel single agent and rarely has been associated with death. In all but these rare cases, patients responded to appropriate medical treatment.

Paclitaxel + Doxorubicin

Congestive heart failure has been reported for combination therapy of Paclitaxel and doxorubicin in previously untreated patients with metastatic breast carcinoma and no prior chemotherapy.

Cases of myocardial infarction have been reported rarely. Cardiac dysfunction and reduction of left ventricular ejection fraction or ventricular failure have been reported typically in patients on Paclitaxel who have received other chemotherapy, notable anthracyclines.

Paclitaxel + Radiotherapy

Radiation pneumonitis has been reported in patients receiving concurrent radiotherapy.

Overdose

At present, there is no specific treatment for Paclitaxel overdosage. Probable consequences of an overdose are mucositis, severe bone marrow suppression and peripheral neurotoxicity. Treatment should be supportive.

Pharmacological properties

Pharmacodynamic properties

Pharmacotherapeutic group: antineoplastic agents (taxanes),

ATC code: L01CD01

Mechanism of action

Paclitaxel is a novel antimicrotubule agent that promotes the assembly of microtubules from tubulin dimers. It stabilizes microtubules by preventing depolymerization resulting in the inhibition of the normal dynamic reorganization of the microtubule network essential for cellular functions. Paclitaxel also induces abnormal arrays or "bundles" of microtubules throughout the cell cycle and multiple asters of microtubules during mitosis.

Pharmacokinetic properties

The pharmacokinetics of paclitaxel have been evaluated over a wide range of doses, up to 300 mg/m², and infusion schedules, ranging from 3 to 24 hours and have been shown to be non-linear and saturable with a disproportionately large increase in C_{max} and AUC with increasing dose accompanied by an apparent dose-related decrease in total body clearance.

Following intravenous administration, paclitaxel exhibits a biphasic decline in plasma concentrations. The initial rapid decline represents distribution to the peripheral compartment and elimination of the drug. The later phase is due, in part, to a relatively slow efflux of paclitaxel from the peripheral compartment. In patients treated with doses of 135 and 175 mg/m² given as 3- and 24-hour infusions, mean terminal half-life has ranged from 13.1 to 52.7 hours, and total body clearance has ranged from 12.2 to 23.8 L/h/m². Mean steady state volume of distribution has ranged from 198 to 688 L/m², indicating extensive extravascular distribution and/or tissue binding. Variability in systemic paclitaxel exposure, as measured by AUC(0-∞) for successive treatment courses is minimal; there is no evidence of accumulation of paclitaxel with multiple treatment courses.

Distribution

On average, 89% of drug is bound to serum proteins; the presence of cimetidine, ranitidine, dexamethasone, or diphenhydramine does not affect protein binding of paclitaxel.

Metabolism

In vitro studies with human liver microsomes and tissue slices showed that paclitaxel was metabolized primarily to 6a-hydroxypaclitaxel by the cytochrome P450 isozyme CYP2C8; and to two minor metabolites, 3-p-hydroxypaclitaxel and 6a, 3-p-dihydroxypaclitaxel by CYP3A4. In vitro, the metabolism of paclitaxel to 6a-hydroxypaclitaxel was inhibited by a number of agents.

Elimination

After intravenous administration of 15-275 mg/m² doses of Paclitaxel as 1-, 6-, or 24-hour infusions, mean values for cumulative urinary recovery of unchanged drug ranged from 1.3% to 12.6% of the dose. This indicates extensive non-renal clearance of paclitaxel. In five patients administered a 225 or 250 mg/m² dose of radiolabeled Paclitaxel as a 3-hour infusion, 14% of the radioactivity was recovered in the urine and 71% was excreted in the feces in 120 hours. Total recovery of radioactivity ranged from 56% to 101% of the dose. Paclitaxel represented a mean of 5% of the administered radioactivity recovered in the feces while metabolites, primarily 6a-hydroxypaclitaxel, accounted for the balance.

Pharmaceutical particulars

List of excipients

Polyoxyl 35 castor oil, Citric acid anhydrous, Dehydrated Alcohol (Ethanol, anhydrous).

Special precautions for storage

Unopened Vials:

Unopened vials of paclitaxel injection are stable until the date indicated on the package when stored below 30°C, in the original package. Neither freezing nor refrigeration adversely affects the stability of the product. If unopened vials are refrigerated, a precipitate may form which redissolves with little or no agitation upon reaching room temperature, product quality is not affected. If the solution remains cloudy or if an insoluble precipitate is noted, the vial should be discarded.

Reconstituted solution :

Solutions for infusion prepared as recommended are stable at ambient temperature (25°C) and lighting conditions for up to 27 hours.

From the microbiological point of view, the product should be used immediately, if not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

Nature and contents of container

A clear tubular glass vial (5ml, 20ml, 50ml) with 20 mm bromobutyl serum coated rubber stopper with 20mm aluminium flip off seals in multidose vials.

Presentation: One vial packed in a carton

Handling and Disposal: Procedures for proper handling and disposal of anticancer drugs should be considered. Care must be taken whenever handling cytostatic products. Always take steps to prevent exposure. This includes appropriate equipment such as wearing gloves and washing hands with soap and water handling such products.

Caution : Cytotoxic agent

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